Technical Hardware Research and Development Plan for VATS (Vehicle Accident Transcript Software)

Regulatory Alignment and System Triggers

To establish a forensic-grade Vehicle Accident Transcript Software (VATS) platform, the underlying hardware must conform to global event data recorder (EDR) regulatory standards. These standards mandate highly specific requirements for collision data collection, physical survivability, and data retrieval methods. For light-duty vehicles with a Gross Vehicle Weight Rating (GVWR) of 3,855 kilograms or less, the system must satisfy the United States Federal Regulation 49 CFR Part 563, which governs the recording of pre-crash and crash parameters, as well as the commercial availability of data retrieval tools. In the European and broader international markets, light-duty passenger and light commercial vehicles are regulated by UNECE Regulation No. 160 (covering M1 and N1 categories), while heavy-duty commercial vehicles, buses, and coaches are governed by UNECE Regulation No. 169 (covering M2, M3, N2, and N3 categories).

The regulatory standards dictate precise recording windows and sampling frequencies. For instance, pre-crash monitoring must capture a continuous data buffer spanning up to 20 seconds before the main crash trigger, while post-crash logging must capture at least 10 seconds of subsequent dynamics. The core parameter for evaluating impact intensity is Delta-V (the cumulative change in velocity). In light-duty applications under 49 CFR Part 563, longitudinal and lateral Delta-V must be recorded starting from crash time zero to 0.25 seconds, sampled at intervals of 0.01 seconds, with the maximum Delta-V logged from crash time zero to 0.3 seconds. Conversely, heavy-duty applications under UNECE R169 require a single-observation Delta-V parameter combined with high-frequency inertial measurements.

To support multiple collision profiles, the VATS device must be capable of recording multi-event crashes. A multi-event crash is defined as the occurrence of two distinct events, the first and last of which begin no more than 5 seconds apart, requiring non-volatile memory architectures capable of maintaining at least five independent event records without risk of overwrite or corruption. System triggers are classified into physical crash thresholds, active safety system interventions, and manual or normal operation trip endings. Under UNECE R169, physical triggers include sudden deceleration, defined as a vehicle speed change of 3.25 meters per second squared (0.33 g) or greater for a continuous duration of 0.7 seconds or more. Dynamic triggers are generated upon the activation of supplemental restraint systems (SRS), antilock braking systems (ABS), emergency brake interventions, or vehicle stability controls. Finally, normal trip completions are recorded under "Last Stop" criteria, which trigger when the vehicle speed remains at zero for 20 seconds after being above 24 kilometers per hour for at least 6 seconds, or when the parking brake is engaged or the master control switch is deactivated.

Operational Criteria	US 49 CFR Part 563	UNECE R160 (Light-Duty)	UNECE R169 (Heavy-Duty)
Applicable Vehicles	Light-duty passenger cars, multipurpose	M1 and N1 category passenger cars and	M2, M3, N2, and N3 category buses,

Operational Criteria	US 49 CFR Part 563	UNECE R160 (Light-Duty)	UNECE R169 (Heavy-Duty)
	vehicles, and trucks under 3,855 kg GVWR	light commercial vans	coaches, and commercial trucks
Mandate Timelines	Voluntary if-equipped standard; enforced on certified vehicles since September 1, 2012	Mandatory for all new vehicle types from July 2022; all new registrations from July 2024	Mandatory for new vehicle types from January 7, 2026; all new registrations from January 7, 2029
Pre-Crash Log Range	5 seconds (with updates expanding towards 20 seconds)	5 seconds of pre-crash parameter tracking	-20 seconds to +10 seconds surrounding the trigger event
Delta-V Sampling	100 Hz sampling (longitudinal & lateral) from 0 to 250 ms	100 Hz sampling from 0 to 250 ms	Single observation metric supported by high-frequency IMU capture
Sudden Deceleration Trigger	Non-reversible restraint deployment or 0.15\text{ g} change within 150 ms	Airbag deployment or physical crash sensor threshold activation	Cumulative speed change \ge 3.25\text{ m/s}^2 (\ge 0.33\text{ g}) for \ge 0.7\text{ sec[span_39](start_span)[span_39](end_span)onds}
Event Buffers	Minimum of 2 events (multi-event capture within 5 seconds)	Dynamic allocation for at least 2 distinct collision events	Non-volatile storage partition dedicated to at least 5 distinct event records
Data Accessibility	Commercially available extraction tools and physical connection methods	Machine-readable, standardized format; extraction blocked without unlocking vehicle	Secure, machine-readable extraction; fully anonymous without linking to personal data

High-Reliability Automotive Power Management Architecture

Automotive electrical networks are highly unstable environments characterized by severe voltage fluctuations and destructive transient events. The primary power path must withstand conduction and coupling disturbances defined by ISO 7637-2, alongside environmental electrical loads specified in ISO 16750-2. Electromagnetic compatibility (EMC) testing under ISO 7637-2 focuses on high-amplitude, short-duration transients (150 nanoseconds to 2 milliseconds) originating from capacitive or inductive coupling, such as Pulse 1 (inductive back-EMF from circuit disconnection reaching -150 V with a decay time of 2 milliseconds). These high-frequency pulses can be successfully clamped using passive components, where a 330 \mu\text{F} input bypass capacitor reduces a -150 V spike to a manageable -16 V.

In contrast, ISO 16750-2 concentrates on long-duration, high-energy voltage excursions from low-impedance sources that cannot be filtered by passive networks. The most severe of these is the alternator load dump, simulated by Test A (unsuppressed) and Test B (suppressed) under clause 4.6.4. Under Test A, if the battery is abruptly disconnected while the alternator is actively supplying current, a voltage surge occurs that can reach peak values (V_P) of 101 V in 12 V systems and 202 V in 24 V systems, with a decay duration (t_d) of up to 400 milliseconds. Other critical ISO 16750-2 test profiles include the engine starting profile (clause 4.6.3, simulating cold-cranking voltage drops down to 4 V), reverse battery conditions (clause 4.7, applying -14 V for 60 seconds), and the superimposed alternating voltage test (clause 4.4, injecting AC ripple up to 2 V RMS on the supply rail).

To guarantee continuous operation throughout these disturbances, the VATS device utilizes an active surge stopper circuit built around the LT4363 high-voltage controller. This controller operates across a wide input range of 4 V to 80 V and resists reverse input transients down to -60 V. The LT4363 controls an external N-channel MOSFET, dynamically regulating its gate voltage to clamp the output voltage to a safe, pre-programmed threshold (such as 25 V or 35 V). In an overcurrent fault, the controller monitors the differential voltage across an inline current sense resistor R_{SNS}. When the voltage across R_{SNS} exceeds the internal current limit threshold of 50 mV, an internal amplifier pulls down on the MOSFET gate, restricting the output current in less than 5 microseconds.

The LT4363 features an adjustable fault timer. During overvoltage or overcurrent events, an internal current source charges a timer capacitor C_{TMR} connected to the TMR pin. If the fault persists and the TMR pin voltage reaches 1.375 V, the gate drive is pulled to ground, turning off the MOSFET to prevent thermal destruction of the pass transistor. The LT4363-1 variant latches off upon timing out and must be reset by pulling the shutdown pin ($\overline{SHDN}$) low for at least 100 microseconds, whereas the LT4363-2 automatically retries operation after a cooling period of approximately 22.8 seconds, operating at an ultra-low retry duty cycle of less than 1% to prevent cumulative thermal buildup.

The reverse working maximum voltage (V_{RWM}) of the input protection elements must be selected to accommodate normal battery voltage variations, which reach up to 16 V in 12 V systems and 32 V in 24 V systems. In unsuppressed networks, the peak clamping current (I_{PP}) and peak pulse power (P_{PP}) of the transient voltage suppressor (TVS) diodes (such as the SM8S24A or SM5S24A series) are determined by the generator's internal resistance (R_i), which ranges from 0.5 Ω to 4 Ω in 12 V systems:

$$I_{PP} = \frac{V_P - V_C}{R_i} \quad P_{PP} = V_C \cdot I_{PP}$$

Design Element	Associated Component	Operational Parameter	ISO Compliance Specification
Transient Voltage Suppressor	SM8S24A TVS Diode	$V_{RWM} = 24.0\text{ V}$ $V_C = 38.9\text{ V}$ at $I_{PP} = 120\text{ A}$	Clamps unsuppressed load dump transients under ISO 16750-2 Test A
Active Surge Clamping	External N-Channel MOSFET	$V_{DS} \ge 100\text{ V}$ $R_{DS(ON)} \le 10\text{ m}\Omega$	Regulates supply rail to a safe output voltage during high-energy surges
Output Regulation Threshold	Resistor Divider (R_2, R_{16})	V_{OUT(cl[span_108](start_span)[span_108](end_span)[span_110](start_span)[span_110](end_span)[span_112](star	Clamps the input to downstream step-down regulators during transient overvoltages

Design Element	Associated Component	Operational Parameter	ISO Compliance Specification
		t_span)[span_112](end_span)amp)} = 35.5\text{ V} V_{FB} = 1.275\text{ V}	
Dynamic Current Limit	Sense Resistor (R_{SNS})	R_{SNS} = 4\text{ m}\Omega V_{SNS(limit)} = 50\text{ mV}	Enforces a fast 12.5\text{ A} current limit in less than 5 \mu\text{s} during short-circuits
Input Bypass Filter	Al-Electrolytic Capacitor (C_{IN})	C_{IN} = 330\,\mu\text{F} / 100\text{ V}	Attenuates high-amplitude coupled pulses under ISO 7637-2 Pulse 1 and Pulse 2a
Fault Shutdown Timing	Timer Capacitor (C_{TMR})	C_{TMR} = 100\text{ nF} t_{fault} = 310\text{ ms}[span_128](start_span)[span_128](end_span) (OV) / 61\text{ ms} (OC)	Dictates the active regulation duration before thermal latch-off or cool-down cycles
Reverse Power Protection	Back-to-Back MOSFETs	V_{SD} \ge -60\text{ V} reverse standoff	Replaces high-loss Schottky diodes to pass ISO 16750-2 clause 4.7
	Overvoltage Inhibit	Resistor Divider (R_{OV4}, R_{OV6})	V_{OV(inhibit)} = 20.0\text{ V}

Mathematical Sizing of the Supercapacitor Backup Energy Storage System

To guarantee that forensic event transcripts are written to non-volatile storage during a catastrophic crash where the main battery is crushed or disconnected, the VATS hardware integrates a supercapacitor backup system. This power-fail energy storage system is managed by the LTC3350 high-current charger and backup controller. During normal operation, the LTC3350 functions as a synchronous buck converter, charging a series stack of supercapacitors with constant-current/constant-voltage (CC/CV) profiles and active cell balancing. The internal balancers eliminate the need for external bleed resistors, adjusting charge imbalances dynamically if cell-to-cell differences exceed 10 mV. Upon power-failure detection at the Power Fail Input (PFI) pin, the LTC3350 switches to a synchronous step-up boost configuration, regulating the system output rail (V_{OUT}) by drawing energy from the supercapacitor stack down to a low cutoff threshold (V_{STK(MIN)}).

Sizing the supercapacitors requires defining the system's electrical load and accounting for end-of-life (EOL) component degradation. Supercapacitor aging is characterized by a decrease in nominal capacitance and an increase in equivalent series resistance (ESR). Typically, EOL is defined as a 30% reduction in nominal capacitance (retaining 70% of initial capacity) and a 100% increase in ESR (doubling the initial value).

The minimum allowable stack voltage at EOL is constrained by the maximum power transfer rule. If the stack voltage falls too low, the internal ESR losses exceed the capacity of the cell to deliver power, defined as:

$$V_{STK(MIN)} \ge 2 \sqrt{P_{BACKUP} \cdot ESR_{stk(EOL)}}$$

The required EOL stack capacitance (C_{EOL}) is calculated using the utilization factor (B), which represents the proportion of stored energy extracted during backup. The utilization factor is defined by the maximum fully charged stack voltage ($V_{STK(MAX)}$) and the minimum operational boost cutoff voltage ($V_{STK(MIN)}$):

$$B = \frac{V_{STK(MAX)} - V_{STK(MIN)}}{V_{STK(MAX)}}$$

Using the LTC3350 mathematical model, the minimum EOL stack capacitance is defined as:

$$C_{EOL} = \frac{4 \cdot P_{BACKUP} \cdot t_{BACKUP}}{\eta \cdot n \cdot V_{CELL(MAX)}^2 \cdot \left[\frac{1+B}{B} + \frac{1-B}{B} \ln\left(\frac{1}{1-B}\right) \right]}$$

where η is the boost converter efficiency, n is the number of series cells, and $V_{CELL(MAX)}$ is the maximum charge voltage per cell. The maximum EOL stack ESR is also limited to prevent excessive voltage drops under load:

$$ESR_{EOL} = \frac{\eta \cdot (1-B) \cdot n \cdot V_{CELL(MAX)}^2}{4 \cdot P_{BACKUP}}$$

To monitor the health of the supercapacitor stack, the LTC3350 utilizes its integrated 14-bit ADC and an internal precision current source to perform automated capacitance and ESR measurements. Once the stack is fully charged, the controller forces the synchronous charger off and pulls a precise current from the top of the stack. The internal timing circuit measures the exact time required for the stack voltage to drop by 200 mV, enabling the internal telemetry processor to calculate the actual capacitance and ESR and report EOL warnings to the host microcontroller over an I^2C/SMBus interface.

Sizing Step	Engineering Parameter	Mathematical Derivation & Model	Computed Design Value
Step 1	Target System Load (P_{BACKUP})	Determined by host MCU write cycles, eMMC write current, and transceivers	5.0 W constant backup power
Step 2	Minimum Backup Duration (t_{BACKUP})	Time required to flush volatile write queues and close LittleFS logs safely	5.0 seconds total hold time
Step 3	Boost Converter Efficiency (η)	LTC3350 synchronous step-up converter efficiency at 5.0 W load	90% converter efficiency ($\eta = 0.90$)
Step 4	Required Boost Input Power (P_{IN})	$P_{IN} = \frac{P_{BACKUP}}{\eta}$	5.5556 W required from stack
Step 5	Series Cell Configuration (n)	Dynamic balance of voltage capability and equivalent stack ESR	$n = 2$ supercapacitors in series
Step 6	Cell Operating Voltage ($V_{CELL(MAX)}$)	Set below absolute limits (2.7 V to 3.0 V) to prolong life	2.5 V per cell ($V_{STK(MAX)} = 5.0\text{ V}$)

Sizing Step	Engineering Parameter	Mathematical Derivation & Model	Computed Design Value
Step 7	Minimum Regulated Cutoff (V_{STK(MIN)})	Minimum operational limit programmed at the LTC3350 PFI pin	2.0\text{ V} minimum input voltage
Step 8	Stack EOL Resistance (ESR_{stk(EOL)})	Worst-case aged stack resistance (2 \cdot ESR_{cell(initial)} \cdot 2.0)	0.12\,\Omega (60\text{ m}\Omega per cell at EOL)
Step 9	Maximum Power Transfer Limit	V_{STK(power)} = 2 \sqrt{P_{IN} \cdot ESR_{stk(EOL)}}	1.6330\text{ V} (Regulated 2.0\text{ V} > 1.6330\text{ V})
Step 10	Usable Energy Utilization (B)	B = \frac{V_{STK(MAX)} - V_{STK(MIN)}}{V_{STK(MAX)}}	60\% energy utilization factor (B = 0.60)
Step 11	Minimum EOL Stack Capacitance	Simulated numeric integration of constant-power discharge profile	2.8300\tex[span_177](start_span)[span_177](end_span)t{ Farads} minimum stack capacity
Step 12	Nominal Stack Capacitance (BOL)	C_{stk(BOL)} = \frac{C_{stk(EOL)}}{0.70} (assumes 70% capacitance retention at EOL)	4.0429\text{ Farads} initial stack capacity
Step 13	Nominal Individual Cell Capacity	C_{cell(BOL)} = C_{stk(BOL)} \cdot 2 (series capacity scaling)	8.0857\text{ Farads} cell capacity
Step 14	Final Toleranced Cell Capacity	C_{final} = \frac{C_{cell(BOL)}}{0.80} (accounts for \pm 20\% part tolerance)	**10.1071\text{ Farads}** (Standard 10\text{ F} cell)

Microcontroller Security Engine, Secure Boot, and Chain of Custody

The VATS primary processing platform is based on the NXP S32K344, an automotive-grade microcontroller integrating a hardware-isolated Hardware Security Engine (HSE). The HSE operates as an autonomous secure enclave, containing its own dedicated processor core, secure flash memory, and hardware cryptographic accelerators for symmetric and asymmetric processing. By isolating all security-critical operations, the HSE prevents key disclosure even if the application code running on the host ARM Cortex-M7 cores is compromised.

```
+--------------------------------------------------------------------------------------+
|                              NXP S32K344 MCU                              |
|                                                                      |
```

```
|   +-----------------------------------+        +-----------------------------------+   |
|   |       Host System Cores          |       |     Hardware Security Engine      |   |
|   |        (ARM Cortex-M7)           |       |         (Secure Enclave)          |   |
|   |                                  |       |                                   |   |
|   |  +-----------------------------+ |       |  +-----------------------------+  |   |
|   |  |  Application Flash Space    | |       |  |     Secure Flash (NVM)      |  |   |
|   |  |  (IVT, Active/Passive App)  | |       |  | (Keys, Certificates, SMRs)  |  |   |
|   |  +--------------+--------------+ |       |  +--------------+--------------+  |   |
|   |                 |                |       |                 |                 |   |
|   +-----------------|----------------+       +-----------------|-----------------+   |
|                     |                                         |                     |
|                     |            Messaging Unit               |                     |
|                     +-------------> [ MU_0 Regs ] <-----------+                     |
|                              (FSR: BOOT_OK, INIT_OK)                                |
+--------------------------------------------------------------------------------------+
```

Platform security begins at device reset via Advanced Secure Boot (ASB). The boot sequence is executed through the following stages:

1. **sBAF Execution:** The hardware-programmed, immutable ROM Boot Assist Flash (sBAF) runs first, initiating low-level device configuration and loading the Image Vector Table (IVT).
2. **HSE Firmware Load:** The sBAF reads the pointer in the IVT, decrypts and authenticates the HSE firmware binary from application flash, and programs it to the dedicated HSE secure code flash.
3. **Partition Verification:** An automatic A/B swap configuration is verified. If active, the system configures secure partition access.
4. **SMR Validation:** The HSE authenticates the application code defined in the Secure Memory Region (SMR) table. The SMR is stored in the secure NVM of the HSE, linking memory boundaries and corresponding signatures (such as RSA or ECDSA).
5. **Core Reset Control:** If the SMR signatures are valid, the HSE processes the Core Reset (CR) table. The CR table controls the release of the host application cores from reset, specifying the target boot address and releasing the host CPU.
6. **Status Assertion:** During the boot sequence, the HSE updates the status bits in the Flash Status Register (FSR) of the Messaging Unit (MU_0). The host application can verify the boot status by reading bits 16 to 31 of the FSR:
   - HSE_STATUS_BOOT_OK (Bit 26): Set to 1 when all pre-boot secure conditions pass.
   - HSE_STATUS_INIT_OK (Bit 24): Set to 1 when the post-boot phase completes successfully, enabling service requests over the MU interface.
   - HSE_STATUS_INSTALL_OK (Bit 25): Set to 1 once the secure key catalogs are formatted.
   - HSE_STATUS_RNG_INIT_OK (Bit 21): Set to 1 when the hardware random number generator is initialized.

To establish an indisputable forensic chain of custody, every event transcript generated by VATS must be signed immediately. The HSE uses the Elliptic Curve Digital Signature Algorithm (ECDSA) with prime curves (such as NIST P-256 or P-384) to generate high-strength, compact signatures that are appended to each log. All private signing keys are generated internally using an AIS31-compliant True Random Number Generator (TRNG) during factory provisioning and

are marked as non-exportable, ensuring they cannot be extracted.
To bind the data to a specific vehicle, the HSE generates a Certificate Signing Request (CSR). This CSR is signed by the internal private key to prove possession, and the OEM's root certificate authority (CA) returns a unique ECU identity certificate structured with a Subject Distinguished Name (DN) combining the internal ECU ID and the vehicle's unique 17-character Vehicle Identification Number (VIN):
\text{Subject DN} = \text{CN=GW01-WVWZZZ3CZWE123456, O=OEM Name, OU=Vehicle PKI, C=DE}

Security Parameter	Hardware Standard	Operational Integration	Forensic Integrity Implication
SMR Signature Verification	ECDSA P-256 with SHA-256 or RSA-3072	Verified by sBAF/HSE before releasing the main application core	Prevents the execution of unauthorized or modified logging firmware
Secure Update Integrity	Dual-Partition A/B Swap Configuration	HSE decrypts passive block, validates signatures, and performs atomic partition swap	Mitigates the risk of bricking during updates and blocks rollback attacks
Forensic Key Generation	FIPS 140-2 / AIS31 Cryptographic TRNG	Internally generates non-exportable ECDSA key pairs inside secure NVM	Protects signing keys from exposure, even under full system teardown
Log Signature Protocol	ECDSA (NIST P-256 / P-384 curves)	Appends a unique cryptographic signature block to every completed transaction	Guarantees non-repudiation and detects data alteration
Access Sanction Controls	Hardware-enforced Sanctions Engine	Disables target cryptographic keys or halts the host CPU on detection of a compromise	Limits key misuse and restricts debug access on compromised hardware
Status Register Tracking	Messaging Unit (MU_0) FSR Register	Continuously monitors status bits: BOOT_OK (Bit 26) and INIT_OK (Bit 24)	Enables the application layer to verify the integrity of the security engine

High-Endurance Physical Storage and Logical Filesystem Architecture

The VATS storage architecture must protect forensic data against physical crash impacts, thermal shock, and write failures. The physical layer utilizes an automotive-grade, JEDEC eMMC v5.1 non-volatile storage chip qualified to AEC-Q100 Grade 2 (supporting an operating temperature range of -40°C to +105°C). To survive extreme temperatures and mechanical vibrations, the eMMC is configured to run in pseudo-Single-Level Cell (pSLC) mode. While standard multi-level cell technologies (MLC or TLC) store multiple bits per cell, pSLC restricts each cell to a single bit. This configuration reduces storage capacity by approximately 50% but yields a threefold increase in write endurance (up to 30,000 program/erase cycles) and improves data retention in high-temperature environments.

At the logical filesystem layer, standard FAT or ext4 filesystems are highly vulnerable to metadata corruption during sudden power failures. To prevent this, the VATS software implements the Little File System (LittleFS Version 2.5), which is structured to handle flash-specific challenges. LittleFS utilizes copy-on-write (COW) semantics. When updating a log file, LittleFS writes the new data to a newly allocated physical block rather than overwriting the active block in-place. The directory pointer is only updated to the new metadata pair via an atomic transaction once the write operation is complete, ensuring that a sudden power disruption during logging results in a safe rollback to the previous valid state.

Copy-on-Write (COW) Logging Flow:

```
Step 1: Locate active metadata block (pointing to current log data).
[Metadata Pointer A] ---> [Active Flash Block 1 (Valid Log Data)]

Step 2: Write new log updates to a newly allocated block (Block 2) while keeping Block 1
untouched.
                    [Active Flash Block 1 (Valid Log Data)]
[Metadata Pointer A]
                    [Allocated Flash Block 2 (New Log Data Draft)]

Step 3: Atomically rewrite and switch the metadata pointer to Block 2 after verification.
                    [Old Flash Block 1 (Marked for Erase/Wear Leveling)]
[Metadata Pointer B] ---> [Active Flash Block 2 (Validated Signed Log)]
```

The interface driver translates logical filesystem operations into eMMC sector commands. LittleFS operates with logical block and offset addresses, which are converted to byte-level offsets and mapped to physical sector block addresses on the JEDEC-compliant eMMC:

$$\text{Physical Address (Sector Block)} = \frac{(\text{Logical Block Number} \cdot \text{Block Size}) + \text{Offset}}{\text{eMMC Sector Size (512 Bytes)}}$$

Storage Layer	Component/Feature	Technical Parameter	Operational Implementation & Purpose
Physical Medium	Automotive eMMC	JEDEC eMMC v5.1	Soldered-down FBGA-153 package to resist intense mechanical shock
Thermal Rating	AEC-Q100 Grade 2	-40°C to $+105^{\circ}\text{C}$ range	Maintains data retention under extreme automotive passenger cabin temperatures
Flash Endurance	pSLC Partitioning	30,000 Program/Erase Cycles	Sacrifices MLC capacity to achieve high reliability and prevent read-disturb errors

Storage Layer	Component/Feature	Technical Parameter	Operational Implementation & Purpose
Filesystem Engine	LittleFS V2.5	13 KB ROM / 4 KB RAM footprint	Designed for microcontrollers to provide crash-resilient storage
Data Integrity	Copy-on-Write (COW)	Atomic metadata updates	Prevents filesystem corruption from mid-write power failures during a crash
Lifespan Tuning	Wear Leveling	Dynamic cycle distribution	Distributes write and erase cycles evenly across flash blocks to prevent premature wear
Defect Mitigation	Bad-Block Remapping	Automatic defect scanning	Scans for failing sectors and remaps data dynamically to healthy flash blocks
Driver Interface	Intermediate Bridge	Logical-to-Physical Block Address translation	Converts 64-bit byte addresses to sector-aligned eMMC command sets

CAN Bus Telemetry and Message Parsing Code Implementation

The VATS platform interfaces with the in-vehicle network via a high-speed Controller Area Network with Flexible Data-Rate (CAN-FD) transceiver. The physical interface consists of a differential two-wire twisted pair terminated with 120-ohm resistors at each bus endpoint. In recessive mode, both CAN High and CAN Low are biased to 2.5 V, resulting in a 0 V differential. In dominant mode, CAN High is driven to 3.75 V while CAN Low drops to 1.25 V, creating a 2.5 V differential to transmit logical bits. CAN-FD accommodates high-rate telemetry payloads by supporting data phase bit-rates up to 8 Mbps.

To secure the in-vehicle CAN-FD bus against spoofing and injection attacks, VATS supports the DBCSec framework. This secure communication mechanism groups electronic control units (ECUs) based on the transmission relationships defined in the system DBC database. Rather than relying on rigid, pre-shared long-term keys, DBCSec utilizes Elliptic Curve Qu-Vanstone (ECQV) implicit certificates and Group Diffie-Hellman (GDH) key exchange to establish secure group keys dynamically during initialization.

At the software layer, the structure of CAN frames differs between vehicle classes. Passenger vehicles utilize 11-bit standard identifiers with proprietary message mappings, whereas commercial and heavy-duty vehicles use the 29-bit extended identifier format defined by SAE J1939, where the identifier is split into a Priority, Parameter Group Number (PGN), and Source Address. Raw data bytes are parsed into physical engineering parameters using the physical scaling formula:

$$\text{Physical Value} = (\text{Raw Value} \times \text{Factor}) + \text{Offset}$$

To avoid the CPU overhead of runtime database parsing, the VATS software toolchain uses tools like dbcc or SignalCandy to compile static DBC files into optimized C code structures and decoding functions during compilation. This code is execution-optimized, utilizing shifts and masks to extract signals and unpack payloads without dynamic memory allocation.

The following C implementation illustrates the structural mapping of typical parsed signals and the execution of a decode sequence:

```c
#include <stdint.h>
#include <stdbool.h>
#include <string.h>

/* Structure representation of decoded vehicle telemetry */
typedef struct {
    float vehicle_speed;      /* Decoded vehicle speed in km/h */
    float engine_speed;       /* Decoded engine RPM */
    float brake_pressure;     /* Decoded brake system pressure in kPa */
    bool brake_switch;        /* Binary brake pedal position state */
} vehicle_telemetry_t;

/* Standard CAN-FD raw frame structure definition */
typedef struct {
    uint32_t can_id;          /* 11-bit or 29-bit identifier */
    uint8_t dlc;              /* Data Length Code (up to 64 bytes for CAN-FD) */
    uint8_t data[64];         /* Frame payload */
} can_fd_frame_t;

/**
 * Parses and decodes vehicle speed from the target CAN frame.
 * Target Message ID: 0x18FEF100 (J1939 Vehicle Speed Message)
 * Signal Parameters: Start bit 8, Length 16 bits, Intel (Little Endian)
 * Scaling Factor: 0.00390625 (1/256), Offset: 0.0, Unit: km/h
 */
bool unpack_j1939_vehicle_speed(vehicle_telemetry_t* telemetry, const can_fd_frame_t*
frame) {
    if (frame->can_id != 0x18FEF100 || frame->dlc < 8) {
        return false;
    }

    /* Extract 16-bit raw value from bytes 1 and 2 (Little Endian) */
    uint16_t raw_speed = (uint16_t)frame->data[1] | ((uint16_t)frame->data[2] << 8);

    /* Apply physical scaling formula */
    telemetry->vehicle_speed = (float)raw_speed * 0.00390625f;
    return true;
}

/**
 * Parses and decodes Engine Speed (RPM).
```

```
 * Target Message ID: 0x0CF00400 (J1939 Electronic Engine Controller)
 * Signal Parameters: Start bit 24, Length 16 bits, Motorola (Big Endian)
 * Scaling Factor: 0.125 (1/8), Offset: 0.0, Unit: rpm
 */
bool unpack_j1939_engine_speed(vehicle_telemetry_t* telemetry, const can_fd_frame_t*
frame) {
    if (frame->can_id != 0x0CF00400 || frame->dlc < 8) {
        return false;
    }

    /* Extract 16-bit raw value from bytes 3 and 4 (Big Endian) */
    uint16_t raw_rpm = ((uint16_t)frame->data[3] << 8) | (uint16_t)frame->data[4];

    /* Apply physical scaling formula */
    telemetry->engine_speed = (float)raw_rpm * 0.125f;
    return true;
}

/**
 * Master frame routing function.
 * Matches input IDs against target PGNs and extracts signals into telemetry.
 */
bool parse_can_fd_frame(vehicle_telemetry_t* telemetry, const can_fd_frame_t* frame) {
    bool speed_parsed = false;
    bool rpm_parsed = false;

    if (frame->can_id == 0x18FEF100) {
        speed_parsed = unpack_j1939_vehicle_speed(telemetry, frame);
    } else if (frame->can_id == 0x0CF00400) {
        rpm_parsed = unpack_j1939_engine_speed(telemetry, frame);
    }

    return (speed_parsed || rpm_parsed);
}
```

Forensic State Estimation and Court Admissibility Framework

To provide an accurate reconstruction of vehicle dynamics leading up to and during an accident, VATS implements a kinematic sensor fusion model. High-rate measurements from a 6-Degree-of-Freedom (6-DOF) micro-electromechanical (MEMS) Inertial Measurement Unit (IMU)—which captures linear acceleration (f) and angular rate (ω)—are integrated with lower-rate GNSS position and CAN wheel speed data. The state vector is modeled in the Error-State Extended Kalman Filter (ES-EKF) framework:

$$x_k = \begin{bmatrix} p_k & v_k & q_k \end{bmatrix}^T \in \mathbb{R}^{10}$$

where p_k is the 3D position vector, v_k is the 3D velocity vector, and q_k is the four-element

unit quaternion representing vehicle orientation.
The state update cycle operates in two phases:

1. Propagation Phase (Prediction)

High-rate IMU measurements are integrated into the non-linear motion model:

$$p_k = p_{k-1} + \Delta t \, v_{k-1} + \frac{\Delta t^2}{2} \left(C_n^s f_{k-1} + g\right) v_k = v_{k-1} + \Delta t \left(C_n^s f_{k-1} + g\right) q_k = q_{k-1} \otimes q\left(\omega_{k-1} \Delta t\right)$$

where C_n^s represents the coordinate transformation matrix from the sensor frame to the navigation frame, and g is the gravity vector.
The error covariance (P_k) is propagated using the system motion Jacobian (F_{k-1}) and noise Jacobian (L_{k-1}), which map the IMU noise variances (σ^2_{acc}, σ^2_{gyro}) over the sample interval:

$$P_k = F_{k-1} P_{k-1} F_{k-1}^T + L_{k-1} Q_{k-1} L_{k-1}^T$$

2. Correction Phase (Measurement Update)

When external measurements (such as GNSS or CAN speed data) become available, the Kalman gain (K_k) is computed, and the error state is calculated:

$$K_k = P_k H_k^T \left(H_k P_k H_k^T + R_k\right)^{-1} x_k = x_k + K_k \left(y_k - h\left(x_k\right)\right) P_k = \left(I - K_k H_k\right) P_k$$

where y_k is the sensor measurement, $h(x_k)$ is the measurement model, and R_k represents the measurement noise covariance.
For event transcripts generated by VATS to be admitted as evidence in U.S. courts, the underlying technology must satisfy the Daubert Standard of admissibility (codified in Federal Rule of Evidence 702). Under the Daubert trilogy (*Daubert*, *Joiner*, and *Kumho Tire*), the trial judge acts as a gatekeeper to ensure that expert testimony and technical data are scientifically valid and relevant.
The admissibility of evidence is evaluated against five key factors:

1. **Falsifiability and Testing:** The technology must be capable of empirical testing and validation. VATS satisfies this by undergoing standardized crash and component shock testing compliant with UNECE R100 Annex 9C and UNECE R169 survivability specifications.
2. **Peer Review:** The underlying theories and methods must be published in peer-reviewed scientific literature. The ES-EKF and kinematic equations utilized by VATS are well-documented in established scientific journals.
3. **Quantifiable Error Rates:** The known or potential error rate of the device must be established. By logging the EKF covariance matrix alongside physical telemetry, VATS provides a mathematically rigorous error bound for every state estimate.
4. **Operational Standards and Controls:** The technology must be operated under consistent standards. VATS ensures operational consistency through rigid, standardized decoding based on compiled DBC files, eliminating interpretation variation.
5. **General Acceptance:** The methodology must have achieved general acceptance within the relevant scientific and engineering community. VATS meets this criterion by complying directly with global automotive EDR standards, including 49 CFR Part 563 and UNECE R160/R169.

To support law enforcement synchronization, the VATS device includes a secure, CJIS-compliant police-sync interface. Using a physically secured extraction port (requiring a

local physical unlock key), the system transmits cryptographically signed event transcripts to authenticated terminals over a secure TLS 1.3 connection. These records are formatted as JSON files containing the signed data payload, the ECDSA signature, and the unique device certificate, allowing automated, secure ingestion into cloud evidence repositories like Axon Evidence while maintaining a complete, auditable chain of custody.
Secure Loca[span_406](start_span)[span_406](end_span)l Extraction and Police-Sync Flow:

```
[VATS Non-Volatile Flash]
      |
      +---> [ECDSA Signed Log File (JSON)]
                |
                v
       (Physical Access Port) ---> Requires physical vehicle unlock key
                |
                v (Encrypted TLS 1.3 
Session)[span_416](start_span)[span_416](end_span)[span_417](start_span)[span_417](end_span)
       [Authenticated Terminal / MDT]
                |
                v (Secure API Upload)
       [CJIS-Compliant Cloud ([span_407](start_span)[span_407](end_span)Axon Evidence)] 
<--- Validates signature & device certificate
```

Forensic Standard	Admissibility Criteria	VATS R&D Design Solution	Admissibility & Forensic Precedent
Federal Rule of Evidence 702	Proponent must show that scientific or technical methods are reliably applied	Auto-logging of EKF 3D state estimations, covariance matrix bounds, and sensor diagnostic flags	*Kumho Tire Co. v. Carmichael* (Extends the gatekeeping role to non-scientific engineering and technical evidence)
Federal Rule of Evidence 707	Machine-generated data must satisfy reliability tests equivalent to expert witnesses	Hardware-enforced cryptographic signing of log records within the NXP S32K3 HSE boundary	Enforces rigorous verification standards for machine-generated and automated evidence
Error Rate Quantification	The technology must have a known or potential error rate	Real-time logging of EKF state covariances (P_k) to establish physical error boundaries	*General Electric Co. v. Joiner* (Limits the admission of data when an excessive analytical gap exists)
Physical Validation	Evidence that the device survives and records crashes accurately	Verification via physical drop-shock and crash testing (UN R100 Annex 9C mechanical shock parameters)	Proves that the structural mounting, wiring, and storage layers maintain data integrity through high-energy impacts

Forensic Standard	Admissibility Criteria	VATS R&D Design Solution	Admissibility & Forensic Precedent
Data Authenticity	Protection against post-crash data manipulation	Log blocks are signed with non-exportable private keys using ECDSA signatures	Prevents "junk science" challenges by verifying that the data has not been modified since generation
Chain of Custody	Verification of data origin and handling	Secure CJIS-compliant local transfer to Axon Evidence cloud via signed certificate verification	Establishes an unbroken, verifiable path from physical crash acquisition to courtroom presentation

Works cited

1. Event Data Recorder (EDR): Black Box for Modern Vehicles - iWave Global, https://iwave-global.com/articles/event-data-recorder-for-modern-vehicles/ 2. 49 CFR Part 563 -- Event Data Recorders - eCFR, https://www.ecfr.gov/current/title-49/subtitle-B/chapter-V/part-563 3. R169e.pdf - UNECE, https://unece.org/sites/default/files/2025-01/R169e.pdf 4. 249 PART 563—EVENT DATA RECORDERS - GovInfo, https://www.govinfo.gov/link/cfr/49/563?link-type=pdf&year=mostrecent 5. EDR: Event Data Recorder - A Simple Intro [UNECE R169] - CSS Electronics, https://www.csselectronics.com/pages/r169-event-data-recorder-edr-intro 6. EDR Compliance Infographics - Sibros Marketing Hub, https://get.sibros.tech/edr-infographics 7. Protecting and Powering Automotive Electronics Systems with No Switching Noise and 99.9% Efficiency, https://www.allaboutcircuits.com/uploads/articles/protecting-and-powering-automotive-electronics-systems_compressed.pdf 8. Automotive Load Dump Protection - Diotec Semiconductor, https://diotec.com/en/application/application-notes/application-note-about-automotive-load-dump-protection.html?file=files/diotec/files/pdf/service/applications/AN_automotive_load_dump_protection.pdf 9. LTspice: Models of ISO 7637-2 & ISO 16750-2 Transients | Analog Devices, https://www.analog.com/en/resources/technical-articles/ltspice-models-of-iso-7637-2-iso-16750-2-transients.html 10. Application Note: Quick Guide on Automotive Load Dump Protection (ISO 16750-2), https://solutions.mccsemi.com/news/application-note-quick-guide-on-automotive-load-dump-protection-iso-16750-2 11. DEMO MANUAL DC1935A - LT4363-1/LT4363-2 High Voltage Surge Stopper with Current Limit, https://www.analog.com/media/en/technical-documentation/user-guides/dc1935af.pdf 12. DC2062A - LT4363-1/LT4363-2 12V Systems ISO-7637-2 Surge Stopper - Verical, https://www.verical.com/datasheet/analog-devices-power-management-development-tools-dc2062a-a-6807228.pdf 13. LT4363 DS – High Voltage Surge Stopper with Current Limit - Analog Devices, https://www.analog.com/media/en/technical-documentation/data-sheets/4363fb.pdf 14. LT4363 High surge stopper shutdown at start up - Electrical Engineering Stack Exchange, https://electronics.stackexchange.com/questions/333671/lt4363-high-surge-stopper-shutdown-at-start-up 15. High Voltage LT4363 Surge Stopper With Current Limit | PDF | Mosfet - Scribd, https://www.scribd.com/document/848440748/High-Voltage-LT4363-Surge-Stopper-With-Current-Limit 16. Load dump protection: Old vs. new ISO standards | EDN - Vishay, https://www.vishay.com/docs/49748/49748.pdf 17. Designing with Supercapacitors | Abracon, https://abracon.com/uploads/resources/Designing-with-Supercapacitors.pdf 18. Supercapacitor

State of Health Measurement with the BQ25856-Q1 - Texas Instruments, https://www.ti.com/lit/pdf/sdaa084 19. LTC 3350 Evaluation Kit | PDF | Capacitor | Resistor - Scribd, https://www.scribd.com/document/327725709/Ltc-3350-Evaluation-Kit 20. LTC3350 (Rev. D) - Analog Devices, https://www.analog.com/media/en/technical-documentation/data-sheets/ltc3350.pdf 21. Make the most of your supercapacitors! Backup controller "LTC3350" - Semiconductor Business -Macnica, https://www.macnica.co.jp/en/business/semiconductor/articles/analog_devices/146152/ 22. Supercap backup circuit provides reliable uninterrupted power - EDN Magazine, https://www.edn.com/supercap-backup-circuit-provides-reliable-uninterrupted-power/ 23. Energy Storage Using Supercapacitors: How Big Is Big Enough? - Analog Devices, https://www.analog.com/en/resources/analog-dialogue/raqs/raq-issue-179.html 24. How to Calculate Supercapacitors for Energy Back Up Applications, https://passive-components.eu/how-to-calculate-supercapacitors-for-energy-back-up-applications/ 25. LTC3350 output power calculation - Q&A - EngineerZone - Analog Devices, https://ez.analog.com/power/f/q-a/563497/ltc3350-output-power-calculation 26. Hardware Security Engine (HSE) - NXP Semiconductors, https://www.nxp.com/applications/technologies/security/automotive-security/hardware-security-engine-hse:HARDWARE-SECURITY-ENGINE 27. S32K3xx Security Overview and Bring Up - Training - NXP Community, https://community.nxp.com/pwmxy87654/attachments/pwmxy87654/S32K/30542/2/12_S32K3xx_Security_Overview_and_Bring_Up_Training.pdf 28. AN13465: S32K3xx Secure Boot - Documentation - NXP Semiconductors, https://docs.nxp.com/bundle/AN13465/page/topics/hse_status.html 29. Guidance Required for Crypto Library and HSE Integration on S32K344 (FRDM-A-S32K344) - NXP Community, https://community.nxp.com/t5/S32K/Guidance-Required-for-Crypto-Library-and-HSE-Integration-on/td-p/2370991/jump-to/first-unread-message 30. Hardware Security Module (HSM) Crypto Key & Signature Management - Embedded systems - Tessolve, https://embedded.tessolve.com/wp-content/uploads/2025/11/Tessolve-HSM_whitepaper6-1.pdf 31. A Blockchain and IPFS-Based Anticounterfeit Traceable Functionality of Car Insurance Claims System - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC10708723/ 32. PKI Suite – Vehicle Certificate Management - Sibros, https://sibros.tech/products/cybersecurity/pki-suite 33. X.509 Certificates in Automotive Security - VxLabs, https://vxlabs.ai/resources/x509-certificates-automotive/ 34. e.MMC for Automotive | AEC-Q100 compliant - ATP Electronics, https://www.atpinc.com/products/automotive-managed-nand-emmc 35. What is AEC-Q100 and it's Specifications? - Infineon Developer Community, https://community.infineon.com/t5/Knowledge-Base-Articles/What-is-AEC-Q100-and-it-s-Specifications/ta-p/248018 36. Silicon Motion's Ferri Family: AEC-Q100 Qualified Embedded Storage, https://www.siliconmotion.com/download/3fs/a/FerriSSD_AEC-Q100_WP_EN.pdf 37. Comparing NAND Flash Technology: SLC, MLC, TLC, and QLC for Industrial Memory Applications - Lexar Enterprise, https://lexarenterprise.com/comparing-nand-flash-slc-mlc-tlc-qlc-industrial-application/ 38. LittleFS - A high-integrity embedded file system | Mbed, https://os.mbed.com/blog/entry/littlefs-high-integrity-embedded-fs/ 39. Little File System (LittleFS) Architecture - Developer Docs - Silicon Labs, https://docs.silabs.com/wifi-developer-guides/latest/memory-littlefs/littlefs-architecture 40. Use LittleFS as SD card file system - NXP Community, https://community.nxp.com/t5/i-MX-RT-Crossover-MCUs-Knowledge/Use-LittleFS-as-SD-card-fil e-system/ta-p/1750205 41. Comparing 4 Popular Embedded File Systems for NOR and NAND Flash - JBLopen, https://www.jblopen.com/comparing-embedded-file-systems/ 42. DBCSec: DBC File-Guided Secure Communication Mechanism for CAN-FD Bus | Request PDF - ResearchGate, https://www.researchgate.net/publication/394181374_DBCSec_DBC_File-Guided_Secure_Communication_Mechanism_for_CAN-FD_Bus 43. jcan CAN Software: DBC Files, CAN-FD & Adapter Support - Industrial Monitor Direct, https://industrialmonitordirect.com/blogs/knowledgebase/jcan-open-source-can-tool-linux-and-windows-configuration-guide 44. How to Decode CAN Bus Data: A Practical Guide for Engineers - Ruptela, https://ruptela.com/decode-can-bus-messages-guide/ 45. How to Decode Vehicle's CAN Bus Data : 8 Steps - Instructables, https://www.instructables.com/How-to-Decode-Vehicles-CAN-Bus-Data/ 46. OBD2 DBC File - Decode Your Car Data - CSS Electronics, https://www.csselectronics.com/pages/obd2-dbc-file 47. CAN DBC File Explained — The Backbone Behind CAN Communication in Automotive | by Koteswara Rao Para | May, 2026 | Medium, https://medium.com/@koteswarao031/can-dbc-file-explained-the-backbone-behind-can-communication-in-automotive-b25db40c1d60 48. Unpack individual signals from CAN FD messages - Simulink - MathWorks, https://www.mathworks.com/help/slrealtime/io_ref/canfdunpack.html 49. howerj/dbcc: CAN DBC to C (and CSV, JSON and XML) compiler using the mpc parser combinator library - GitHub, https://github.com/howerj/dbcc 50. CANdb++ .dbc to C/C++ .h converter - EEVblog, https://www.eevblog.com/forum/microcontrollers/vector-candb-to-cc-h-converter/ 51. InitusNovus/Signal-CANdy: DBC to C Code Generator written in F# · GitHub, https://github.com/InitusNovus/Signal-CANdy 52. Robust Vehicle Pose Estimation Through Multi-Sensor Fusion of Camera, IMU, and GPS Using LSTM and Kalman Filter - MDPI, https://www.mdpi.com/2076-3417/15/22/11863 53. jasleon/Vehicle-State-Estimation: This extended Kalman filter combines IMU, GNSS, and LIDAR measurements to localize a vehicle using data from the CARLA simulator. - GitHub, https://github.com/jasleon/Vehicle-State-Estimation 54. Daubert Standard | Wex | US Law | LII / Legal Information Institute, https://www.law.cornell.edu/wex/daubert_standard 55. Daubert standard - Wikipedia, https://en.wikipedia.org/wiki/Daubert_standard 56. Daubert in Detail: How the Admissibility Standard for Expert Testimony in Court Has Evolved, https://www.faro.com/en/Resource-Library/Article/Daubert-in-Detail 57. Law 101: Legal Guide for the Forensic Expert | Daubert and Kumho Decisions, https://nij.ojp.gov/nij-hosted-online-training-courses/law-101-legal-guide-forensic-expert/pretrial/pretrial-rules-evidence/daubert-and-kumho-decisions 58. Secure, AI-Powered Digital Evidence Management - Axon.com, https://www.axon.com/products/axon-evidence